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StudentLife Stress Coach

Personalized support with language agents

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The coaching task

Suggest
Ask
Stay silent

A useful message must fit the student's situation and preferences.

StudentLife observations

- Inferred sleep, activity and social indicators
- EMA stress and mood
- Academic schedules and deadlines
- Pre-term survey background

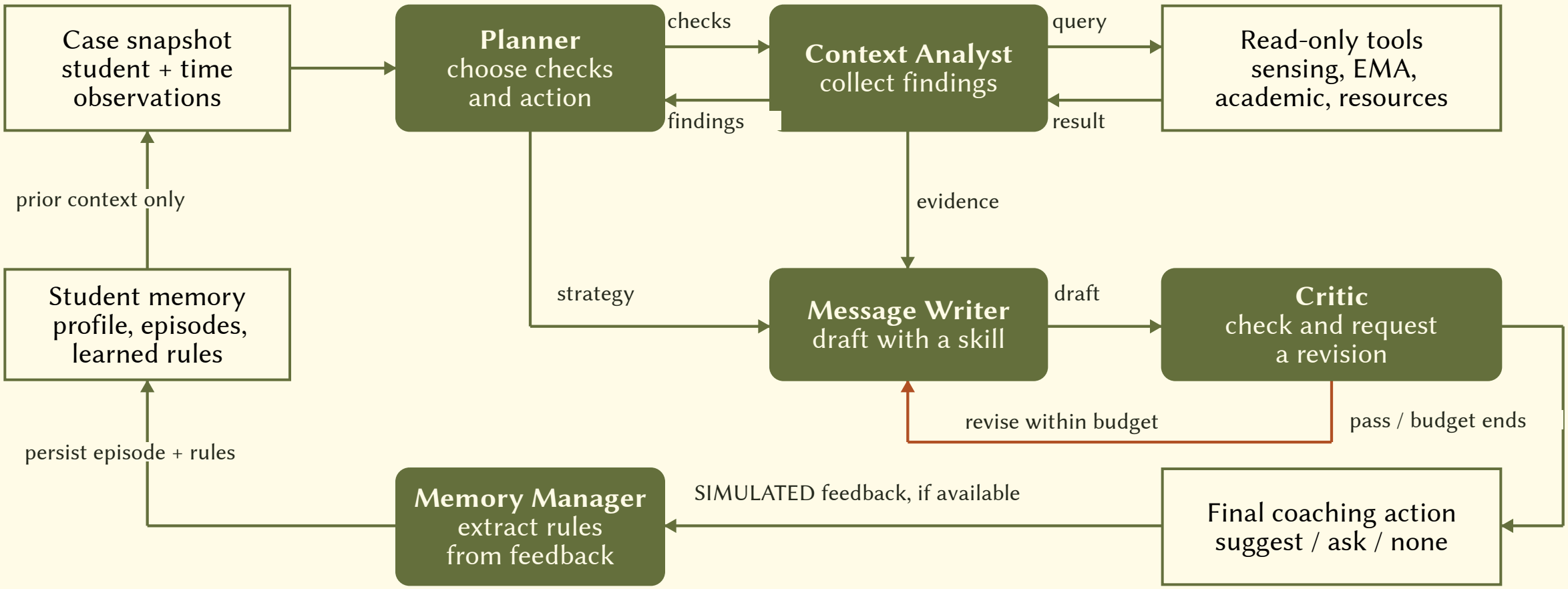
Clearly labeled simulation

Goals, coaching preferences and scripted feedback fill information the dataset does not contain.

20 cases across 8 participants

49 identifiers in the supplied activity archive. The published study describes 48 students over ten weeks.

Five interacting roles



Simplified full configuration. Silence bypasses writing/review. Feedback follows the action. JSONL serialization includes the memory update.

Five required design patterns

Reflection	Critic reviews the draft and requests revisions. Maximum 2 cycles.
Tool use	Analyst chooses data queries. Writer can load a coaching skill.
Planning	Planner selects evidence checks and the action. Maximum 2 calls.
Multi-agent system	Five roles exchange structured, validated outputs.
Memory and skills	JSON history and learned rules, plus 8 reusable Markdown procedures.

Full system: Gemini 3.6 Flash. Independent judge: OpenAI GPT-5.4 mini.

All patterns can be switched or varied in TOML. Full retains five recent episodes and prior rules. No training or fine-tuning.

Memory across interactions

Student u16

April 15
Simulated reply prefers quiet breathing, stretching or tea.

April 18
New simulated feedback requests library-compatible messages under 35 words.

April 22
The next interaction can use both earlier updates.

April 22: a deadline and reported stress

The coach selects `micro_study_break`.

“Midterms and today's deadline can feel overwhelming. Take a quick 5-minute quiet pause right at your desk to sip warm tea and do a few slow, deep breaths.”

28 words. Quiet, desk-compatible support.

Actual full-system output, case_015. Profile preferences and replies are simulated. The baseline already favors quiet breaks, so this example does not isolate a causal memory benefit.

Full-system results

Human rates 10 messages before seeing the independent judge's scores.

20 / 20

Agreement with allowed actions

4.81 / 5

Mean quality across 16 messages

13 suggestions, 3 questions,
4 silent decisions.

Human and LLM scores

Criterion	Human	Judge
Context awareness	4.70	4.80
Goal alignment	4.80	4.90
Clarity	5.00	4.60

0.267 mean absolute difference

30 criterion ratings. Largest gap: clarity.

Mean efficiency: 35.59 seconds, 6.20 LLM calls, 17,529 tokens per case.

Rubric: context awareness, goal alignment, clarity (1-5). Silence has no quality score. Latency includes coaching/tools and excludes judging.

Ablation tradeoffs

Nine configurations use the same 20 case definitions. All produce 20 valid final outputs.

Configuration	Quality (1-5)	Allowed actions	LLM calls/case (0-8)
Full	4.81	20 / 20	6.2
Single coach	4.81	20 / 20	3.65
No planning	4.71	20 / 20	6.1
No reflection	4.78	18 / 20	5.35
Planning max. 1	4.78	19 / 20	5.4
Planning max. 3	4.87	19 / 20	6.95
Reflection max. 1	4.63	20 / 20	6.25
Reflection max. 3	4.83	20 / 20	6.1
Model B	4.89	19 / 20	5.8

The single coach matches rounded quality with fewer calls.

Larger budgets do not consistently improve results. Model B scores 4.89, with one fewer allowed decision.

Conditional quality: 15-16 messages. Histories and compute can differ. One generation per condition.

Conclusions

Five roles make plans, evidence and remembered preferences inspectable.

More roles and larger budgets did not consistently improve message quality.

Small offline benchmark, simulated goals and feedback, and one human rater. We do not measure changes in student stress.

Mahdi Moghaddami
Data, tools, orchestration and experiments. Human ratings.

Brett Piggott
Writing/critique, memory, skills and automated evaluation. Figures and report.

Appendix: all nine conditions

Configuration	Quality	Agreement	Calls	Tokens (k)	Latency (s)
Full	4.81	100%	6.2	17.53	35.59
Single coach	4.81	100%	3.65	18.68	18.78
No planning	4.71	100%	6.1	18.37	49.63
No reflection	4.78	90%	5.35	15.98	68.54
Planning max. 1	4.78	95%	5.4	15.3	46.27
Planning max. 3	4.87	95%	6.95	20.37	80.79
Reflection max. 1	4.63	100%	6.25	17.92	46.38
Reflection max. 3	4.83	100%	6.1	17.35	52.94
Model B	4.89	95%	5.8	14.36	21.6

Depth finding: reflection maxima 1, 2 and 3 each execute one review per sent message on average. Actual Planner calls average 1, 2 and 2.05 at maxima 1, 2 and 3.

All values are means. Conditional quality cohorts contain 15-16 messages. Latency includes provider waits and recovery work. Each condition has its own evolving history.

Appendix: tools and coaching skills

Eight data and resource tools

- `get_time_context`
- `get_sleep_summary`
- `get_activity_summary`
- `get_social_summary`
- `get_stress_mood_history`
- `get_academic_load`
- `compare_to_baseline`
- `search_campus_resources`

Writer: `load_skill`. Single coach also has `recall_memory`.

Eight reusable skill procedures

- `micro_study_break`
- `sleep_hygiene_nudge`
- `gentle_movement`
- `social_reconnection`
- `deadline_chunking`
- `campus_resource_referral`
- `clarifying_question`
- `positive_reinforcement`

Skill files provide applicability guidance, constraints and writing instructions.

Data tools expose bounded observations and explicit missing-data status. Campus resources are manually curated. Tool windows default to three days where applicable.

Appendix: models, input and output

Exact model settings

Model A: full and structural ablations

`gemini:gemini-3.6-flash`

Temperature 0.3, Critic/Memory Manager 0.1. Thinking low. 8,192 output tokens.

Model B: same full architecture

`gemini:gemini-3.5-flash-lite`

Same role temperatures and output cap.

Independent judge

`openai:gpt-5.4-mini-2026-03-17`

Temperature 0, reasoning none, 2,048 completion tokens.

Participant at a decision time

Input: observations available by that time, a simulated profile, and strictly earlier interactions.

One final JSONL record per case


```
{
  "case_id": "case_015",
  "participant_id": "u16",
  "decision": "suggest",
  "skill_used": "micro_study_break",
  "trace": {
    "llm_calls": 7,
    "total_tokens": 19326,
    "latency_s": 26.44
  }
}
```


Projected actual record. Full output also stores evidence, frozen context, plans, critiques and memory updates.